Although the formula has a branching structure, a maximum operation can be easily avoided when implementing it in code. The quantile loss in Python looks like this:

```
def quantile_loss(q, y, y_hat_q):
    """ Calculate the quantile loss for a given quantile.

    Args:
    q (float): The quantile to be evaluated, e.g., 0.5 for median.
    y (float): The target value.
    y_hat_q (float): The quantile forecast.
    """
    error = y - y_hat_q
    return np.maximum(q * error, (q - 1) * error)
```

Now, let's get into the practical side of things and learn how to use quantile loss with the different techniques (both machine learning and deep learning) we have covered in this book.

Forecasting with quantile loss (machine learning)

Regular machine learning models are typically capable of modeling one output. Therefore, we will need to train a different model for each of the quantiles we are interested in using quantile loss. So, if we want to predict the 0.5, 0.05, and 0.95 quantiles for a problem, we will have to train three separate models, one for each quantile.

Notebook alert:

To follow along with the complete code, use the notebook named `04-LightGBM_Prediction_Interval_Quantile_Loss.ipynb` in the `Chapter17` folder.

Let's see how we can do that. Just like in the PDF section, we are using `mlforecast` to quickly whip up a synthetic problem and create some features. For the detailed code, refer to the full notebook, but for now, let's assume that we have a dataframe `data`, which has all the features necessary to run a machine learning model. We have split it into `train` and `val` and then subsequently into `X_train`, `y_train`, `X_val`, and `y_val`.

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